DO NOW — RECALL

RAM, ROM and virtual storage

Name: _____ Class: _____ Date: _____

1. Why are SSDs and HDDs physically different?

2. What purpose does processor cache serve?

3. Which is normally volatile: RAM or ROM?

4. Sorted array [2,6,10,14,18,22,26,30]. Find 22 with midpoint indexes and range updates.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

SSDs use semiconductor cells; HDDs use rotating magnetic platters.

2. Model answer

Keeps some needed data/instructions in fast nearby memory.

3. Model answer

RAM is volatile.

4. Model answer

low0 high7 mid3→14; low4 high7 mid5→22, found at index 5.

Vocabulary — know these terms

RAM — Volatile read/write working memory.

ROM — Non-volatile firmware/boot storage.

virtual storage — Remote network-accessible file storage.

virtual memory — OS technique using address translation and backing storage.

firmware — Low-level code retained without power.

NEW LEARNING

Read, annotate and explain

RAM versus ROM

RAM supports active execution and is volatile; ROM and firmware storage retain instructions without power.

Define virtual storage

A remote or cloud storage service keeps files on network-accessible systems, potentially available on multiple devices.

Distinguish virtual memory

OS virtual memory involves memory addressing/paging with secondary storage, taught again in OS memory management.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Why can unsaved editing work disappear after power loss while firmware remains?

Predict

Unsaved temporary program state depends on volatile RAM and can disappear without power.

Explain

Firmware persists in non-volatile storage, allowing startup instructions to remain available.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State typical use and whether RAM and ROM are volatile.

Question 2

Explain two benefits and one drawback of network-based virtual storage for pupil coursework.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Distinguish virtual storage from virtual memory and explain one limitation of each.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Virtual cloud storage is another name for OS virtual memory.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.